

Homework 6

1. Short answer questions

- (a) Consider a cell in a contingency table. Given the cell's row total of 70, the cell's column total of 75, and a sample size of 250, what is the cell's expected frequency under hypothesis of independence of row and column attributes?
- (b) A chi-square test of independence is applied to a contingency table with 5 rows and 6 columns for two qualitative variables. What will be the degrees of freedom for this test?
- (c) A chi-square goodness-of-fit test is always conducted as: upper tailed, lower tailed, two-tailed. Choose the correct answer.

2. Consider the data used in class on distribution of opinions about marijuana possession before and after television series has aired. The distribution before was Decriminalization (7%), Legalization(18%), Existing law(65%), No Opinion (10%). After the series aired, 500 individuals were randomly selected and interviewed. Their opinions were recorded as Decriminalization (39), Legalization(99), Existing law(336), No Opinion (26).

- (a) Set this up as a hypothesis testing problem, specifying the null and alternative hypotheses, parameter, data and test statistic.
- (b) Use the result that the distribution of the test statistic is asymptotically chi-squared, to obtain the p-value, stating the appropriate degrees of freedom.
- (c) The function
`multinom_test`
in the R-library `rstatix` carries out an exact test in this situation. Find the p-value using this method.
- (d) The exact test takes a long time and the asymptotic test is approximate. Another possibility is to generate random vectors from the multinomial distribution under the null hypothesis using the function `rmultinom`, computing the test statistic of this random vector and simulating the distribution of the test statistic under the null hypothesis. Use a sample of size 10^5 to obtain the p-value using this method.

3. A social scientist wants to determine whether the marital status (divorced or not divorced) of U.S. men is independent of their religious affiliation (A-D) or lack thereof. A sample of 500 U.S. men is surveyed, and the results are tabulated as shown in the following table.

	A	B	C	D	None	Totals
Divorced	39	19	12	28	18	116
Married, never divorced	172	61	44	70	37	384
Totals	211	80	56	98	55	500

Test to see whether there is sufficient evidence to indicate that the marital status of men who have been or are currently married is dependent on religious affiliation. Take $\alpha = .01$.